

General Tasks (25 points)

In addition to the case study, you must complete several general coding tasks in a Jupyter Notebook. Briefly explain each step of your data analysis and support your results with visualizations. Partial points may be awarded for incomplete solutions if the documentation is clear. (Please run all code before saving so that the latest outputs are visible. The code must also run with the latest version of Anaconda.)

The first general task is to identify a suitable probability distribution for the calculated data. Research appropriate statistical goodness-of-fit tests and apply suitable Python methods independently.

1. Logistics and Product Development in the Automobile Industry (8 Points)

Logistics plays an increasingly important role in the product development of the automobile industry. Parts produced by suppliers must first be delivered to the OEM (Original Equipment Manufacturer) before they can be installed. While this process may seem straightforward, a detailed analysis is necessary for professional application.

Create a distribution for the logistics delay of component "K7." Use the production date ("Produktionsdatum") from the dataset "Komponente_K7.csv" and the receiving date of incoming goods ("Wareneingang") from "Logistikverzug_K7.csv" (logistics delay). Assume that produced goods are issued one day after the production date. For the model design in Python, create a new dataset "Logistics delay" containing the required information from both datasets.

- **a.** How is the logistics delay distributed? Justify your choice with statistical tests and briefly describe your approach. (2 points)
- **b.** Determine the mean logistics delay, in terms of working days (Weekend days are excluded). Interpret this number and discuss possible alternatives. (2 points)
- **c.** Visualize the distribution appropriately by displaying the histogram and density function using "plotly." Describe how you selected the size of the bins. (2 points)
- **d.** Describe the process for creating a decision tree model to classify whether the component (K7) is defective (Fehlerhaft) or not. (Hint: Use visualizations.) (2 points)

2. Data Storage in Separate Files (2 Points)

- Explain why it makes sense to store the available data in separate files instead of saving everything in one large table. Name at least four benefits. The available tables represent a typical database structure. What is this structure called?

3. Parts T16 in Registered Vehicles (3 Points)

- Determine how many parts T16 ended up in vehicles registered in Adelshofen.

4. Attributes of the Registration Table (2 Points)

- Identify the data types of the attributes in the registration table "Zulassungen_aller_Fahrzeuge." Present your answers in a table integrated into your Markdown document and describe the characteristics of the data types.

5. Linear Model for Mileage (5 Points)

- Create a linear model from the table “Fahrzeuge_OEM1_Typ11_Fehleranalyse” relating mileage to suitable variables. Derive recommendations for OEM1 based on this model.

6. Hit and Run Accident Investigation (5 Points)

- On 11.08.2010, there was a hit-and-run accident. The license plate of the car involved is unknown. The police have asked for your assistance, as you work for the Federal Motor Transport Authority, to find out where the vehicle with body part number “K5-112-1122-79” was registered.